Inequality in Life Expectancy Across Small Areas in the United States and Colombia

Natalia Serna

Corresponding author, E-mail: nserna@stanford.edu, Phone number: 608-215-7841, Affiliation: Department of Health Policy, Stanford University, Address: 615 Crothers Way, Stanford, CA, 94305

Iván Mejía-Guevara

E-mail: gauss75@gmail.com, Affiliation: Research Consultant, Palo Alto, CA, 94305

Paul Rodríguez-Lesmes

E-mail: paul.rodriguez@urosario.edu.co, Affiliation: Department of Economics, Rosario University, Bogotá, Colombia

Eran Bendavid

E-mail: ebd@stanford.edu, Affiliation: Department of Medicine, Stanford University, Stanford, CA, 94305

Grant Miller

E-mail: ngmiller@stanford.edu, Affiliation: Department of Health Policy, Stanford University, Stanford, CA, 94305

Abstract

Background

Inequality in life expectancy within high-income countries has received considerable attention in the last decade, but much less is known about this topic within lower-income countries. This paper compares inequality in life expectancy at birth in Colombia, a country with unusually rich small-area census data, with similar measures of inequality in the U.S.

Methods

We used publicly available census tract-level data from Colombia's 2018 population census, which report population counts and deaths by five-year age groups as well as measures of economic position. For comparison, we draw on existing census tract-level estimates of life expectancy at birth for the U.S. produced by the Centers for Disease Control and Prevention. We estimated age-specific mortality rates for each Colombian census tract using the TOPALS (Tool for Projecting Age-Specific Rates Using Linear Splines) relational model and assessed the sensitivity of the estimates to alternative mortality models (D-Spline and Log-Quadratic), alternative cluster sizes, and the exclusion of uncertified deaths. Differences between Colombia and the U.S. were assessed using 95% confidence intervals computed by simulation from the asymptotic distribution of the TOPALS parameters. Our primary outcome is life expectancy at birth, which we compute by applying standard demographic methods to the estimated age-specific mortality schedules.

Results

There is a moderate gap of 2 to 5 years in average life expectancy at birth between the U.S. and Colombia, but Colombians in the bottom 1% of the life expectancy distribution live 15 years less than their American counterparts. We investigate the role of income in explaining this inequality in life expectancy, finding that a 10-percentile increase in income rank was associated with 0.32 years of additional life expectancy in the U.S. relative to Colombia. Moreover, census tracts with life expectancy at birth below 50 years were present in every income decile in Colombia.

Conclusions

Our results suggest that factors other than income play a larger role in longevity in Colombia (and potentially in other lower-income countries like Colombia) than they do in the U.S., underscoring the importance of broader social determinants in shaping health outcomes in lower-income countries.

Keywords: Life expectancy, Household income, Social determinants of health

Background

Within high-income countries, a robust body of research documents substantial inequality in life expectancy, with income playing a central role [9, 15, 16, 18]. For example, in the United States (U.S.), the difference in life expectancy at age 40 between the top and bottom one percent of the income distribution is 15 years for men and 10 years for women [8]. This income gradient in life expectancy has drawn considerable attention in public health and policy circles, pointing to critical social determinants of health.

In contrast, far less is known about inequality in life expectancy in low- and middle-income countries (LMICs). Existing research has examined cross- and within-country differences in life expectancy with a focus on individual cities or age groups [5, 20]. But the shape and dispersion of socioeconomic gradients in life expectancy across the population remain poorly characterized in these settings.

Understanding this relationship is especially important in LMICs, where income inequality is often substantially greater than in high-income countries. In Brazil and Chile, for example, earnings in the top decile of the income distribution are roughly 30 times higher than those of the bottom half of the population – a disparity nearly three times larger than in the U.S. [7]. Such extreme income inequality raises the possibility that socioeconomic gradients in longevity may differ in both magnitude and shape from those documented in high-income countries. Whether they do carries direct policy stakes: if income strongly determines survival, then individual transfers and tax benefits or government transfers to municipalities are plausible instruments for narrowing longevity gaps across individuals and places, respectively; if not, such policies may prove ineffective, and interventions addressing broader social determinants of health may be warranted, particularly in LMICs marked by violence and internal displacement [24].

This study addresses this gap in the literature by comparing within-country inequality in life expectancy at birth, and its income gradient, between a middle-income country and the U.S. The focus is on Colombia, whose 2018 population census provided, for the first time, census tract-level data on both socioeconomic status and age-specific mortality. Colombia combines high income inequality with universal health insurance [11] – opposite to the U.S.

– making it well-suited for testing whether the income-longevity gradient documented in the U.S. generalizes to lower-income settings.

In this study we aimed to estimate life expectancy at birth for each census tract in Colombia; compare the resulting distribution with that of the U.S.; and investigate how differences within and between countries vary across the income distribution.

Methods

Data and Statistical Methods for Colombia

We used publicly available data from the 2018 Colombian population census, provided by the National Administrative Department of Statistics (DANE), covering all 504,996 census blocks (*manzanas*) in Colombia's 1,121 municipalities (see https://microdatos.dane.gov.co/index.php/catalog/643/get-microdata). For each census block, the data report population and deaths in 2017 by sex and 5-year age group, as well as socioeconomic status (SES), a 0-6 classification of residential properties assigned by municipal governments to assess utility fees and subsidies, based on housing characteristics and access to public services.

We grouped census blocks within each of Colombia's 33 states into clusters comparable in size to U.S. census tracts using K-means clustering on geographic centroid coordinates and SES decile (see details in Supplementary Material 1). We set the number of blocks within each cluster to 100.

For each resulting cluster, hereafter "census tract," we estimated 5-year age-specific mortality schedules using the Tool for Projecting Age-Specific Rates Using Linear Splines (TOPALS) relational model [12], which fits a linear spline to the ratios of observed age-specific mortality rates to a standard schedule. Estimation proceeded in four nested steps: state, urban/rural area, socioeconomic decile, and census tract. Each step's estimates served as the standard schedule for the next, initialized with the national schedule from the 2024 United Nations (UN) World Population Prospects (see https://population.un.org/wpp/downloads?folder=Standard%20Projections&group=Mortality). We then computed

life tables and life expectancy at birth using standard demographic methods. We performed clustering and mortality estimation in R (version 4.3.1; R Foundation for Statistical Computing, Vienna, Austria) and the remaining analyses in Stata (version 18; StataCorp LLC, College Station, TX). Supplementary Material 1 provides details.

Sensitivity analyses. We conducted several analyses to assess the robustness of the Colombian estimates. Estimated life expectancy distributions showed accurate in- and out-of-sample fit (Supplementary Material 3: Figures S4–S8). Results were similar after excluding uncertified deaths (Supplementary Material 4: Figure S10) and using clusters of 50 or 200 census blocks rather than 100 (Supplementary Material 5: Figure S11). Life expectancy estimates were also similar when the TOPALS relational model was replaced with a D-Spline model [22] or a Log-Quadratic model [26] (Supplementary material 6), although the Log-Quadratic model yielded higher mortality rates at ages 65 and older, likely reflecting its calibration to data from high-income countries. Finally, 95% CIs computed by simulation from the asymptotic distribution of the TOPALS parameters showed that the long left tail of the Colombian distribution was not attributable to estimation uncertainty.

Data and Statistical Methods for the United States

For the U.S., we obtained census tract-level estimates of life expectancy at birth for 2010–2015 from the U.S. Small Area Life Expectancy Estimates Project (USALEEP) (see https://www.cdc.gov/nchs/data-visualization/life-expectancy/index.html) and matched them to 2011–2015 American Community Survey estimates of median household income at the census tract level (see https://data2.nhgis.org/main). Household income is defined as total pre-tax money income received over the previous 12 months by the householder and all other household members aged 15 years or older. Because USALEEP estimates are already reported at the tract level, no clustering was required.

Measures of Economic Position

Because SES is unavailable for U.S. census tracts and household income is unavailable for most Colombian census tracts, we assessed the comparability of the two measures in Bogotá, where both exist. Bogotá represents approximately 20% of the Colombian population and 10% of all census blocks. Household income in Bogotá comes from DANE's 2011 National Quality of Life Survey and is measured as pre-tax monetary income received in the past month by all household members aged 12 or older, annualized to match the U.S. measure. In Bogotá, SES and household income are strongly positively correlated, have similar dispersion, and produce similar gradients in life expectancy at birth (Supplementary Material 2: Figure S2, Panels A–C), supporting their use as comparable markers of economic position.

Statistical Analysis

To compare the gradients of life expectancy with respect to economic position across countries, we pooled the data from both countries and regressed mean life expectancy at birth on SES/income percentile, an indicator for the U.S., and their interaction. The interaction coefficient measures the difference in slopes between the two countries, and we report it with its standard error. We also report the R^2 from country-specific regressions of life expectancy at birth on SES (Colombia) or household income (U.S.) across census tracts. Finally, we quantified uncertainty in the Colombian estimates with 95% confidence intervals obtained by drawing 1,000 parameter vectors from the asymptotic normal distribution of the TOPALS estimates, recomputing life expectancy at birth for each draw, and taking the 2.5th and 97.5th percentiles of the resulting distribution.

Results

Of Colombia's 504,996 census blocks, 397,381 had nonmissing SES data; the mean block population was 92 residents, and 76% of blocks recorded zero deaths (Table 1). Mortality rates estimated at this scale would be dominated by sampling noise, since a single death in a block of 92 residents implies a crude death rate more than ten times the national average.

Aggregating blocks into larger units before estimation addresses this problem. Implementing our K-means clustering approach yielded 4,127 clusters with a mean population of 8,846 and a mean of 42 deaths, reducing the proportion of units with zero deaths to 1%. The U.S. comprised 63,828 census tracts with a mean population of 4,442. Mean life expectancy at birth was 78.4 years in the U.S. vs 74.1 years in Colombia, with less dispersion across geographic units in the U.S. (SD, 3.95 vs 6.50 years in Colombia).

We observed substantial variation in life expectancy at birth across census tracts within the largest cities of each country (Figure 1). Life expectancy at birth ranged from 41 to 84 years in Bogotá and from 59 to 94 years in New York. In Bogotá, census tracts with the lowest life expectancy (dark red) were concentrated in the urban periphery. We observed similar patterns in other major cities in both countries (Barranquilla, Bucaramanga, and Manizales in Colombia; Houston, Phoenix, and Philadelphia in the U.S.; Supplementary Material 5: Figure S12). The within-city variation points to local determinants of mortality that operate below the national or regional levels. Supplementary Material 5, Figure S13 shows maps for entire countries.

The distribution of life expectancy at birth across census tracts differed markedly between countries. The distribution for Colombia (red) was shifted left relative to the U.S. (white) and exhibited a longer left tail, including census tracts with life expectancy at birth below 50 years (Figure 2, Panel A). Low life expectancies in Colombia were not concentrated in rural census tracts; the left tail of the distribution was similar in urban and rural areas (Supplementary Figure S14, Panel A).

However, unlike the U.S., these low life expectancies could be attributed to Colombia's long history of internal armed conflict and drug trade wars, which have displaced large populations into urban areas and depressed rural ones. Between 1948 and 1958, Colombia experienced a period of partisan violence between Liberals and Conservatives (*La Violencia*) that gave rise to the rural insurgencies which consolidated as the FARC and ELN guerrillas in the 1960s [21]. Violence surged again in the 1980s with the rise of drug cartels and paramilitary groups [19]. The main paramilitary federation demobilized between 2003 and 2006, and the FARC demobilized following the 2016 peace agreement, although the ELN and dissident groups remain active. To investigate whether this history of violence can explain

the low life expectancies we observe, we use geolocated records of terrorist attacks from the *Centro Nacional de Memoria Histórica* and classify a census tract as exposed if its centroid lies within 2 kilometers of an attack. Consistent with this explanation, the distribution of life expectancy at birth among exposed census tracts is shifted to the left relative to that among unexposed tracts (Supplementary Material 5: Figure S15).

Life expectancy by percentile of the within-country distribution was flatter in the U.S. than in Colombia (Figure 2, Panel B): the gap between the top and bottom percentiles was 20 years in the U.S. vs 34 years in Colombia. The between-country gap in life expectancy at birth also varied sharply across the distribution (Panel C). For the top 90% of the distribution, the difference between the two countries was 5 years or less; in the left tail, the gap widened rapidly, reaching 15 years [95% CI, 13.2–16.4] at the bottom percentile.

Life Expectancy by Measures of Economic Position

Because measures of economic position vary across census tracts, they are natural candidates for explaining the small-area geographic variation in life expectancy at birth. Our findings show that the association of life expectancy at birth with household income was steeper in the U.S. than the association with SES in Colombia (Figure 3). A 1-percentile increase of a census tract's position in the income distribution was associated with 0.032 (SE, 0.005) additional years of life in the U.S. relative to Colombia. The difference in gradients was not attributable to the Colombian SES measure being coarser than household income, nor to Colombian census tracts being more rural than U.S. tracts. First, the U.S.–Bogotá difference in gradients, which compares household income in both settings, exceeded the U.S.–Colombia difference (Supplementary Material 2: Figure S3), which is the opposite of what a coarseness artifact would produce. Second, restricting Colombia to urban census tracts yielded an SES gradient in life expectancy at birth of similar magnitude to that for all census tracts (Supplementary Material 5: Figure S14, Panel B).

Figure 4 presents the distribution of life expectancy at birth by household income decile in the U.S. and by SES decile in Colombia. In Colombia, median life expectancy increased weakly with SES, and the gradient was driven mainly by census tracts in the lower range of each decile (Panel A). Even conditional on SES decile, the distribution retained a long

left tail: census tracts with low life expectancy were present in all deciles. In the U.S., by contrast, gains across income deciles occurred throughout the within-decile distribution, and within-decile variation was smaller than in Colombia (Panel B). Household income explained 36% of the variation in life expectancy across census tracts in the U.S. while SES explained only 7% in Colombia (R^2 from country-specific regressions).

Panel C compares the life expectancy gap distributions of the two countries separately for the top (blue) and bottom (red) income/SES deciles. To construct this figure, we first ranked census tracts in the bottom income/SES decile within each country into percentiles of life expectancy at birth. Then, at each percentile of the life expectancy distribution, we computed the difference between the U.S. and Colombian values. We repeated the same procedure for census tracts in the top income/SES decile.

Two patterns stand out. Between-country differences in life expectancy at birth were largest at the bottom of the life expectancy distribution across income/SES deciles. At the first percentile of life expectancy, census tracts in the U.S outlived Colombian census tracts by 10 years [95% CI, 6.6–12.6] in the red line and by and 28 years [95% CI, 23.8–32.1] in the blue line; by the 20th percentile these gaps had fallen to roughly 0 and 4 years, respectively. Second, the U.S. advantage is systematically larger in the top income decile than in the bottom decile, and in the bottom decile it even reverses: for example, at the 90th percentile of the life expectancy distribution, low-SES Colombian tracts outlived low-income U.S. tracts by 3 years [95% CI, 2.6–3.3].

Discussion

This study compared the distribution of life expectancy at birth across census tracts in Colombia and the U.S., and its gradient with respect to measures of economic position (SES in Colombia, household income in the U.S.). Our findings show that the distribution of life expectancy at birth had a much longer left tail in Colombia than in the U.S. Colombian census tracts with life expectancy below 50 years were present in every SES decile, suggesting SES plays a small role in survival inequalities. Consistent with this, economic position explained far less of the variation in life expectancy in Colombia than in the U.S. (7% vs 36% of

the variance across census tracts), and the gradient was flatter. A 10-percentile increase in income rank was associated with 0.32 fewer additional years of life expectancy in Colombia than in the U.S.

This study makes two main contributions. First, it shows that comparisons of mean life expectancy either across countries [14] or within countries [5, 13] may miss important disparities present in other parts of the life expectancy distribution [4]. The finding that the gap in first-percentile life expectancy at birth between Colombia and the U.S. was constant across income groups suggests that policies promoting upward economic mobility alone, such as income transfers or tax deductions, are unlikely to reduce the cross-country gap (consistent with prior work [3, 10]).

If the pattern of long left tail in life expectancy observed in Colombia is also present in other lower-income countries after controlling for income, this means that other factors may play a larger role in health inequality compared to the U.S. [23], such as violent deaths and forced migration [1, 2, 6, 17]. Policies focused on public safety and reintegration of displaced populations may thus deserve attention as health policy in such settings, pointing to an important direction for future research on the determinants of mortality in lower-income countries.

Second, small-area life expectancy estimates permit examination of the income-longevity association for the entire population rather than only for individuals reporting income through the public finance system. For example, studies in high-income countries document substantial disparities in life expectancy at age 40 by income because mortality data can typically be linked to income records only for these age groups [8, 15]. However, income does not influence survival only in adulthood [25].

Although the focus on Colombia and the U.S. was motivated by differences in economic inequality and the availability of small-area mortality records, these findings may extend to other settings with high economic inequality. Countries characterized by substantial economic informality, younger populations, and exposure to conflict [24] may exhibit an income-longevity relationship similar to Colombia's. Improving survival in such settings may require moving beyond policies targeting economic inequality alone and toward those addressing broader social determinants of health.

This study has limitations. First, the Colombian socioeconomic measure is not identical to U.S. household income; however, where both measures were available, they were highly correlated and produced indistinguishable life expectancy gradients. Second, the estimates for the two countries cover different periods and were constructed using different methods. Yet, the cross-country contrasts of interest are large and remain stable across a range of sensitivity analyses in Colombia, making it unlikely that they are attributable to these methodological differences. Finally, findings from a single middle-income country may not generalize to all low- and middle-income settings, although the mechanisms examined, such as economic informality, violence, and internal displacement, are common to many.

Conclusions

In this cross-sectional study, the gap in average life expectancy at birth between Colombia and the United States was modest, but it reached 15 years at the bottom of the life expectancy distribution and persisted across all income levels. Income was a substantially weaker predictor of survival in Colombia than in the U.S. These findings suggest that reducing survival disparities in lower-income countries may require policies that address broader social determinants of health rather than income alone.

Declarations

Ethics approval and consent to participate. Not applicable.

Consent for publication. Not applicable.

Availability of data and materials. All the datasets generated and/or analyzed during in this study are available in the Harvard Dataverse https://doi.org/10.7910/DVN/BMDFWG

Competing interests. The authors declare that they have no competing interests.

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Authors' contributions. N.S. and G.M. conceived and designed the study. P.R.-L. and N.S. assembled and curated the data. N.S. and I.M.-G. performed the statistical analyses. N.S. drafted the manuscript. E.B., G.M., I.M.-G., and P.R.-L. revised the manuscript for important intellectual content. G.M. supervised the project. All authors read and approved the final manuscript.

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Additional Files

Additional file 1: Supplementary Materials. Supplementary methods, figures, and tables (PDF). Contains: (i) methodological details of the K-means clustering of census blocks and the TOPALS estimation; (ii) a comparison of socioeconomic status and household income as measures of economic position in Bogotá; (iii) in-sample and out-of-sample fit of the mortality model by urban/rural area and SES decile; (iv) a comparison of census-based and vital-statistics-based life expectancy estimates and results excluding uncertified deaths; (v) robustness of the main results to alternative cluster sizes, urban/rural stratification, and exposure to terrorism, together with country-level maps and maps of additional cities; and (vi) re-estimation of life expectancy with the D-Spline and Log-Quadratic mortality models.

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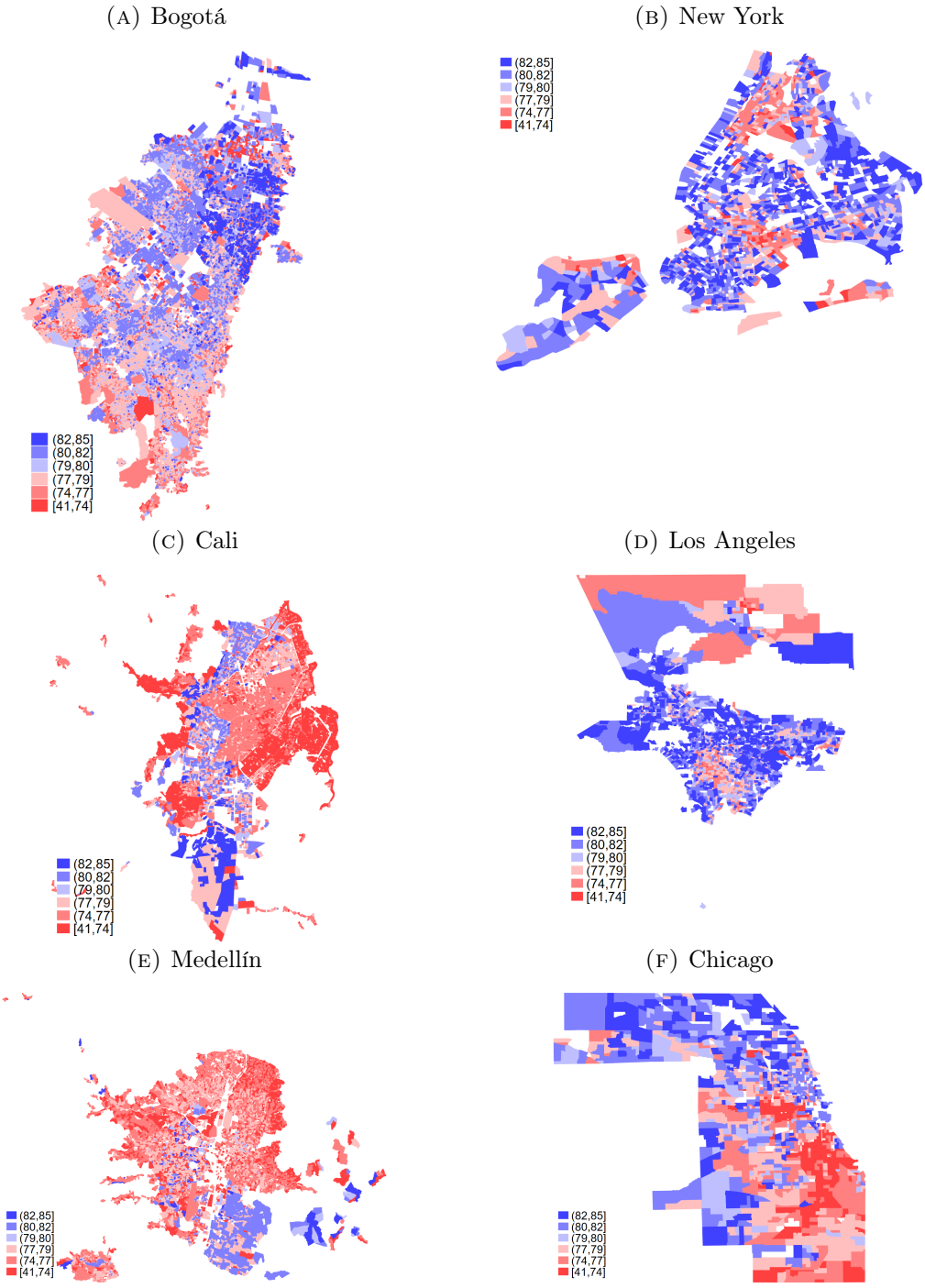
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TABLE 1: Summary Statistics

Variable	Colombia		USA
	Census blocks	Clusters	Census tracts
	(1)	(2)	(3)
Life expectancy	74.31	74.10	78.40
	(5.91)	(6.50)	(3.95)
Number of deaths	0.44	42.00	—
	(1.55)	(47.01)	
Fraction with zero deaths	0.76	0.01	—
	(0.43)	(0.12)	
SES	1.79	1.76	—
	(0.91)	(0.95)	
Median per capita income (1000s)	—	—	57.86
			(28.34)
Total population	91.87	8,845.71	4,442.35
	(139.44)	(9,912.42)	(2,063.55)
Number of census blocks	397,381	96.29	
		(74.25)	
Number of census tracts		4,127	63,828

Note: Table presents the mean and standard deviation in parenthesis of the main variables in our study. In column (1) an observation is a Colombian census block conditional on those with non-missing SES information. In column (2) an observation is a cluster of Colombian census blocks, which we denote in the main text as census tract. In column (3) an observation is a. U.S. census tract. Total U.S. census tract population is obtained from the 5-year American Community Survey estimates for 2011-2015. Socieconomic status (SES) information is not available for U.S. census tracts and median per capita income is not available for Colombia census blocks.

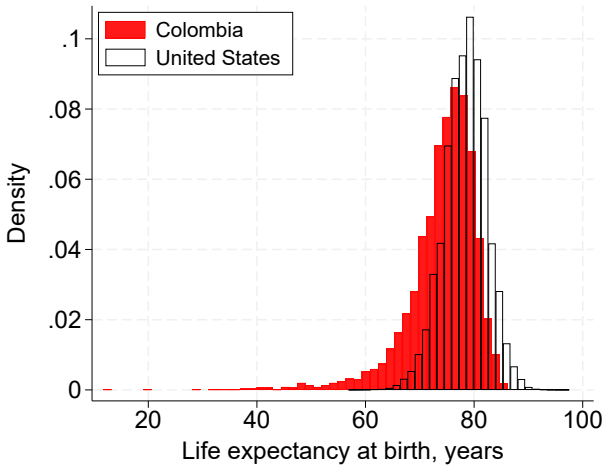
FIGURE 1: Geographic Distribution of Life Expectancy at Birth in Colombia and the United States



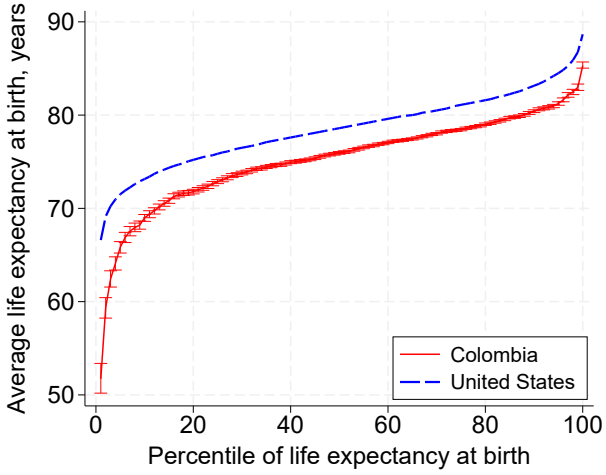
Note: Figure presents the geographic distribution of life expectancy at birth in Colombia for the cities of Bogotá, Cali and Medellín (Panels A, C, E respectively) and in the U.S. for the cities of New York, Los Angeles, and Chicago (Panels B, D, F, respectively). Darker blues represent higher values and darker reds represent lower values.

FIGURE 2: Distribution of Life Expectancy at Birth in Colombia and the United States

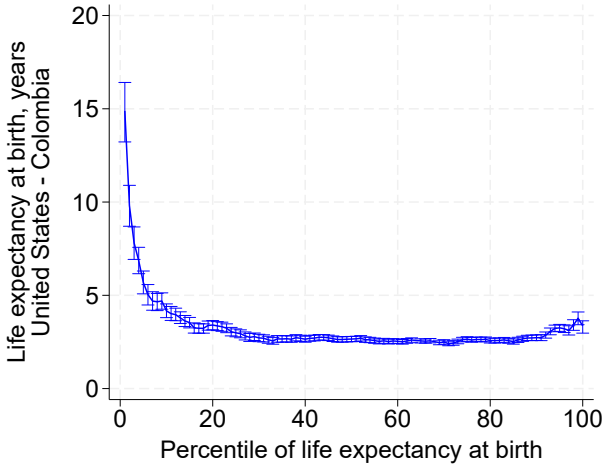
(A) Distribution of life expectancy at birth



(B) Mean life expectancy at birth by percentile

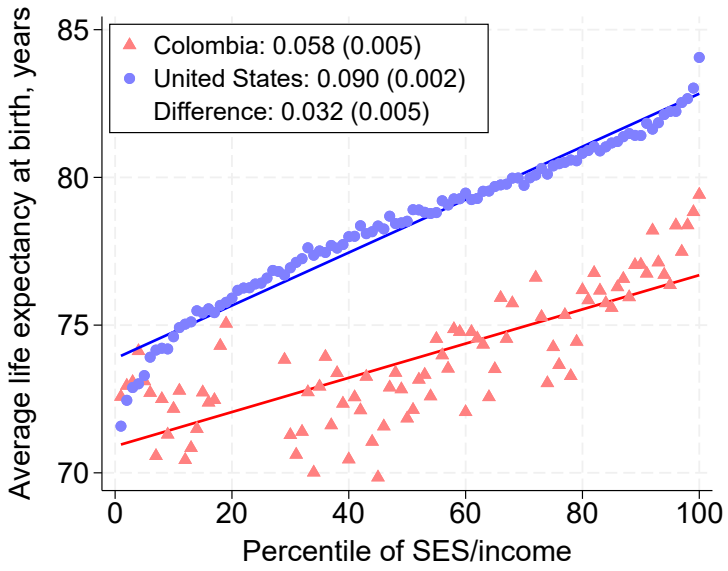


(C) Between-country gap in life expectancy at birth



Note: Panel A presents the distribution of life expectancy at birth across census tracts in Colombia (red) and in the U.S. (white). Panel B presents the mean life expectancy at birth by percentile of the distribution in Colombia (red) and the U.S. (gray). Panel C presents the gap in mean life expectancy at birth between the U.S. and Colombia for census tracts at each percentile of the distribution of life expectancy at birth. 95% confidence intervals reflect uncertainty in Colombian estimates only and were computed by simulation.

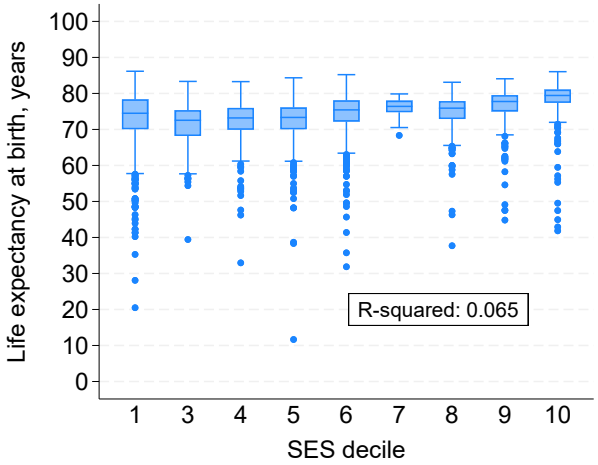
FIGURE 3: Mean Life Expectancy by SES/Income Decile in Colombia and the United States



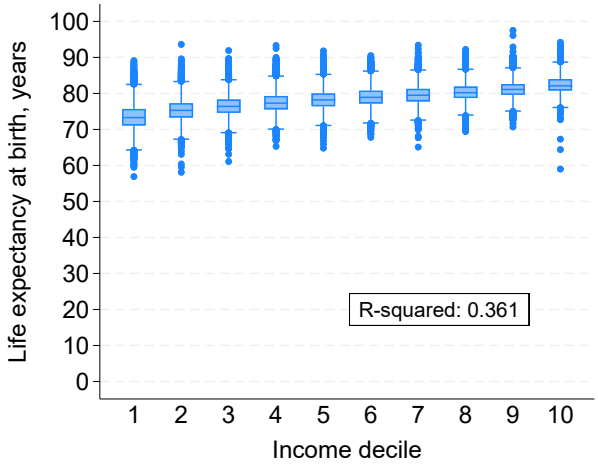
Note: Figure presents mean life expectancy at birth in Colombia (red) by percentile of socioeconomic status (SES) and in the U.S. (gray) by percentile of household income. Straight lines correspond to linear fits. Socioeconomic status in Colombia is a 0-6 categorization of residential units with the purpose of charging differential utilities fees. The text box in the figure reports the coefficient and standard error in parenthesis of a linear regression of mean life expectancy at birth on SES/household income percentile estimated separately for each country. The text box also reports the difference in slopes between the two countries and its standard error in parenthesis. To compute the difference in slopes, we estimate a linear regression pooling the data from the two countries. The dependent variable is mean life expectancy at birth, and the independent variables are SES/income percentile, a dummy variable for the U.S., and the interaction between SES/income percentile and the U.S. dummy variable. The difference in slopes is the coefficient on this interaction.

FIGURE 4: Mortality and Life Expectancy by Income in Colombia and the United States

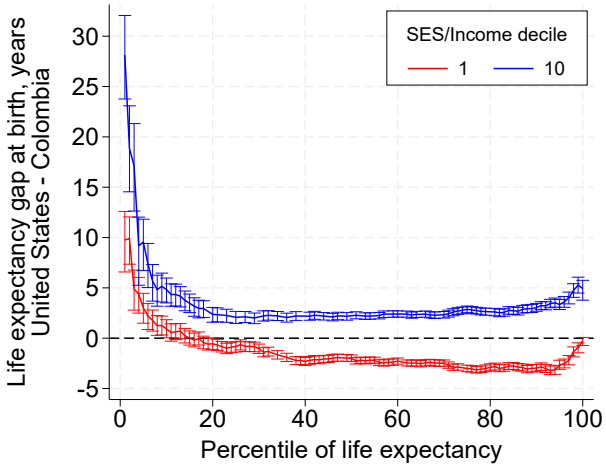
(A) Life expectancy distribution by Socioeconomic Status in Colombia



(B) Life expectancy distribution by income in the United States



(C) Life expectancy gap at birth between the United States and Colombia by economic position



Note: Panels A and B present the distribution of life expectancy at birth conditional on deciles of socioeconomic status (SES) in Colombia and on deciles of median household income in the U.S., respectively. The center line in each box represents the median of the distribution. The bottom and top of each box represent the 25th and 75th percentiles, respectively. The bottom whisker is the 25th percentile minus 1.5 times the interquartile range. The upper whisker is the 75th percentile plus 1.5 times the interquartile range. Socioeconomic status in Colombia is a 0-6 categorization of residential units with the purpose of charging differential utilities fees. Each panel reports the R-squared of a linear regression of life expectancy at birth on median household income in the U.S. and on socioeconomic status in Colombia. An observation in these regressions is a census tract. Panel C presents the gap between the U.S. and Colombia at every percentile of the distribution of life expectancy at birth conditional on top and bottom income deciles. 95% confidence intervals reflect uncertainty in Colombian estimates only and were computed by simulation.